

Project Initialization and Planning Phase

Date	03 July 2026
Team ID	XXXX
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop an AI-powered crop recommendation system that predicts the most suitable crop based on soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. The system helps farmers make informed decisions, improve crop productivity, and reduce financial risk using Machine Learning.
Scope	The project focuses on crop prediction using a Random Forest Machine Learning model trained on agricultural datasets. It provides crop recommendations through an easy-to-use web application. Future enhancements include fertilizer recommendation, weather integration, explainable AI (SHAP), and yield prediction.
Problem Statement	
Description	Farmers often select crops based on traditional knowledge rather than scientific analysis. Improper crop selection reduces productivity, increases farming costs, and leads to financial losses. There is a need for an intelligent decision-support system that recommends the best crop using environmental and soil parameters.
Impact	The proposed system improves crop selection accuracy, increases agricultural productivity, reduces farming risks, and supports sustainable farming practices by enabling data-driven decisions.
Proposed Solution	
Approach	Develop a web-based Machine Learning application using a Random Forest classifier. The application accepts soil nutrient values (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall as inputs and predicts the most suitable crop. Predictions are generated instantly and displayed through an interactive user interface.
Key Feature	AI-based Crop Recommendation Random Forest Prediction Model User-friendly Web Interface Real-time Prediction Soil Parameter Analysis

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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Intel Core i5/i7 (or equivalent AMD processor)	Intel Core i5 / AMD Ryzen 5
Memory	8 GB RAM (Minimum)	8GB
Storage	10 GB Free SSD Storage	10 GB SSD
Software		
Frameworks	Flask	
Library	Scikit-learn matplotlib Pandas joblib Numpy pickle	
Development Environment	Visual Studio Code , Jupyter Notebook / Google Colab	
Data		
Kaggle Crop Recommendation Dataset	Approximately 2200 records	CSV